



Answers & Solutions:

1. **D. all of the above**
2. **D. reduced by one-quarter.**
3. **B. 86.5 %**
4. **A. the instant in the voltage cycle at which circuit is closed**
5. **B. $X_L = X_C$**
6. **A. 39.35 V**
7. **B. One time constant.**
8. **C. narrower bandwidth.**
9. **B. I_L is less than I_C .**
10. **D. Becomes capacitive**
11. **C. capacitive.**
12. **A. 0.309 μF**

Solution:

$$\tau = RC = 15 \text{ ms}$$

$$C = \frac{\tau}{R} = \frac{15 \times 10^{-3}}{50 \times 10^3} = 0.309 \mu\text{F} \rightarrow \text{Ans.}$$

13. **A. a higher L/C ratio.**
14. **A. 0° .**
15. **D. It is at maximum**
16. **A. resistive**
17. **D. resistance, inductance and capacitance**
18. **C. minimum.**
19. **A. 17 kHz**

Solution:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi\sqrt{(88 \times 10^{-3})(1000 \times 10^{-12})}}$$

$$f_r = 16.965 \text{ kHz} \rightarrow \text{Ans.}$$

20. **D. 500 mV_{p-p}.**

Solution:

At resonance

$$V_C = QV_{in}$$

$$V_C = (100)(5 \text{ mV})$$

$$V_C = 500 \text{ mV}_{p-p} \rightarrow \text{Ans.}$$

21. **C. have no stored energy**

22. **C. capacitive.**

23. **C. Current in the circuit to build up to 63.2% of the maximum value.**

24. **C. 23.53 s**

Solution:

When the voltage falls to 25 V,

$$25 = 120e^{-\frac{t}{15}}$$

$$\frac{25}{120} = e^{-\frac{t}{15}}$$

$$-\frac{t}{15} = \ln\left(\frac{25}{120}\right)$$

$$t = -15 \ln\left(\frac{25}{120}\right)$$

$$t = 23.53 \text{ s} \rightarrow \text{Ans.}$$

25. **B. Undamped**

26. **A. decrease.**

27. **A. 7.07 mA_{p-p}**

Solution:

$$I_{\text{HPF}} = 0.707 I_{\text{max}}$$

$$I_{\text{HPF}} = 0.707(10 \text{ mA}_{p-p})$$

$$I_{\text{HPF}} = 7.07 \text{ mA}_{p-p} \rightarrow \text{Ans.}$$

28. **D. Both A and B**

29. **A. resistive**

30. **B. maximum.**

31. **C. inductive**

32. **A. 55.90 μF**

Solution:

When discharging, capacitor voltage,

$$v_C = V e^{-\frac{t}{CR}}$$

$$v_C = \frac{2}{3} V \text{ when } t = 17\text{s}$$

$$\frac{2}{3} V = V e^{-\frac{17}{C(750 \times 10^3)}}$$

$$\frac{2}{3} = e^{-\frac{17}{C(750 \times 10^3)}}$$

$$\ln\left(\frac{2}{3}\right) = -\frac{17}{C(750 \times 10^3)}$$

$$C = 55.90 \mu\text{F} \rightarrow \text{Ans.}$$

33. **D. 63.2% of the supply voltage.**

34. **A. minimum.**

35. **C. 9.45 pF**

Solution:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$C = \frac{1}{(2\pi f_r)^2 L}$$

$$C = \frac{1}{(2\pi)^2 (14.1 \times 10^6)^2 (13.5 \times 10^{-6})}$$

$$C = 9.45 \text{ pF} \rightarrow \text{Ans.}$$

36. **C. 13.5 %**

37. **A. 3.49 MHz**

Solution:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi\sqrt{(4.44 \times 10^{-6})(470 \times 10^{-12})}}$$

$$f_r = 3.48 \text{ MHz} \rightarrow \text{Ans.}$$

38. **D. capacitive**



39. **B. 15/4 sec**

Solution:

$$\text{Effective resistance} = \frac{5 \times 3}{5 + 3} = \frac{15}{8} \Omega$$

$$\begin{aligned} \text{Time constant} &= RC = \left(\frac{15}{8}\right)(2) \\ &= \frac{15}{4} \text{ sec} \rightarrow \text{Ans.} \end{aligned}$$

40. **A. maximum**

41. **B. is highly selective**

42. **C. Increased shunt resistor**

43. **D. Both A and B.**

44. **A. One time constant.**

45. **D. maximum.**

46. **C. decrease to half**

47. **B. the flywheel effect.**

48. **B. increase by a factor of two**

49. **B. minimum**

50. **B. transition**

END